Maximilien Le Cleï

Lead Applied Research Scientist @ D-BOX Technologies | PhD student @ Mila

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Professional Experience

Lead Applied Research Scientist @ D-BOX Technologies, Toronto (Remote), Canada — Jan 2024 - Jun 2025 + Jul 2026 - Present

- Convinced management and succesfully led the company in applying modern SOTA to imitate D-BOX haptic artist tracks (generative modeling of 1D signals – chair actuators displacement – with audio+video conditional diffusion transformers | state space models)
- Strategized with leadership, given modeling successes, to create previously infeasible solution offerings predicted to significantly expand D-BOX's ~40M dollar revenue generating customer base, with generous newfound profit margins given AI-centricity
- Training billion parameter models on terabytes of modern D-BOX data. Running carefully-crafted, systematic scaling experiments over task complexity, training regimes, architectural choices, input modalities, model capacities & dataset sizes
- PhD research application: continual learning & out-of-distribution generalization, now building on 20+ years of D-BOX craftsmanship

PhD Student Researcher @ Mila - Quebec AI Institute, Toronto (Remote), Canada — Oct 2024 - Mar 2027 (est.)

- Attempting to evolve beyong the AI paradigm: rawer human behaviour modeling through the expansion of random space exploration
- Human agentic & brain behaviour imitation (focus: continual learning & out-of-distribution generalization), GECCO 2024 publication

Research Scientist @ Montreal Geriatric Institute, Toronto (Remote), Canada — Sep 2021 - Aug 2023

- Led the development of a strongly automated and highly collaborative machine learning (deep learning + evolutionary machine learning) training workspace to speed up and strengthen the experiments of present machine learning researchers
- Technical contribution: audio model alignment with fMRI data to improve generalization, published in Imaging Neuroscience 2025
- Technical advising: large language model (LLM) fine-tuning + visual language model (VLM) for fMRI brain data alignment
- Evolutionary machine learning for human video game behaviour cloning & modern reinforcement learning benchmarks
- + work leading to publications. First author: ICLR ALOE workshop 2022, GECCO 2022 & 2023; 2 x Imaging Neuroscience 2026

MSc Student Researcher @ Mila - Quebec AI Institute, Montréal, Canada — Jan 2019 - Aug 2021

- Research and development of novel evolutionary machine learning algorithms for imitation and reinforcement learning
- Developed i) a highly scalable and distributed open-source general purpose evolutionary machine learning library ii) a novel high-level imitation learning paradigm and iii) extremely compute efficient artificial neural networks

Applied Research Scientist Intern @ Capbeast, Montréal, Canada — Jan 2018 - Apr 2018

- Early-stage application of modern deep reinforcement learning techniques to industrial embroidery problems
- Developed embroidery virtual environments of various complexity/fidelity & applied state-of-the-art deep reinforcement learning algorithms (DQN, Prioritized experience replay & Dueling Double-Q-Learning) to prototype highly capable embroidering agents

Publications

1. Y Harel, B Pinsard, JA Boyle, A Cyr, **M Le Cleï**, P Mignot, M St-Laurent, K Jerbi, P Bellec. "Gamer in the scanner: Event-related analysis of fMRI activity during retro videogame play guided by automated annotations of game content." *Imaging Neuroscience*, 2026.
2. M Freteault, **M Le Cleï**, L Tetrel, L Bellec, N Farrugia. "Alignment of auditory artificial networks with massive individual fMRI brain data leads to generalisable improvements in brain encoding and downstream tasks." *Imaging Neuroscience*, 2025.
3. A Kemtur, F Paugam, B Pinsard, Y Harel, P Sainath, **M Le Cleï**, J Boyle, K Jerbi, P Bellec. "Behavioral imitation with artificial neural networks leads to personalized models of brain dynamics during videogame play." *Imaging Neuroscience*, 2025.
4. **M Le Cleï**, S Bar-Sheshet, P Bellec. "Generational Information Transfer with Neuroevolution on Control Tasks." *GECCO*, 2024.
5. **M Le Cleï**, P Bellec. "Generative Adversarial Neuroevolution for Control Behaviour Imitation." *GECCO*, 2023.
6. **M Le Cleï**, P Bellec. "Neuroevolution of recurrent architectures on control tasks." *GECCO*, 2022.

Education

University of Montréal, Montréal, Canada

Doctor of Philosophy - PhD, Computer Science (Artificial Intelligence) — Sep 2023 - Mar 2027 (est.)

Master of Science - MSc, Computer science (Artificial Intelligence) — Sep 2018 - Aug 2021

Concordia University, Montréal, Canada

Bachelor of Computer Science - BCompSc, Computer Science (Computer Systems) — Sep 2015 - Apr 2018

Software

Research software: maximilienleclei/artificial_sapience_research (paradigm evolution attempt: raw human behaviour + random space exploration), courtois-neuromod/cneuromax (collaborative research workspace), maximilienleclei/nevo (evolutionary machine learning)
Large public repositories: google/evojax (scalable evolutionary machine learning), ROCm/TheRock (AMD ROCm/PyTorch)